Data-driven Wildfire Frequency Prediction for Real-Estate Applications

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Motivation

We aimed to address two foundational issues in traditional wildfire simulations:

High computational cost:

Physical fire simulations require large computing resources

Static assumptions:

Historical ignition and fuel maps do not adapt to

- climate change
- urban expansion
- changing vegetation patterns

Objective

To develop a scalable, data driven approach for estimating wildfire risk across California

Requirements

Insurance is a high-stakes application requiring models to be ▸ Interpretable ▸ Reproducable

Data

PRISM (*Climate*)

▸ Precipitation ▸ Dewpoint temperature
▸ Max temperature ▸ Vapor pressure deficit

NVDI (*Normalized Difference Vegetation Index*)

▸ Vegetation density and moisture
▸ Proxy for available wildfire fuel

NLCD (*Terrain Classification*)
Classifies terrain into 52 categories including:

▸Forest ▸Shrubland ▸Grassland ▸Urban

USGS DEM (*Topography/Digital elevation data*)

▸ Elevation ▸ Slope ▸ Aspect

MTBS (Burn Severity)
Historical wildfire perimeters sourced from Cal Fire Trap
▸Used to generate binary labels

Predictors

Target

Data Proccessing

PRISM monthly climate layers are clipped to California and used as the spatial reference for the rest of the pipeline.

MODIS NDVI is regridded to the PRISM reference so vegetation values line up with the climate cells.

Annual land-cover maps are upscaled to the common grid and repeated across months within each year.

Slope, and aspect are derived from the DEM and aggregated to the same analysis grid.

California wildfire perimeters are filtered by year and rasterized by ignition month to create monthly labels.

Results

Performance Metrics

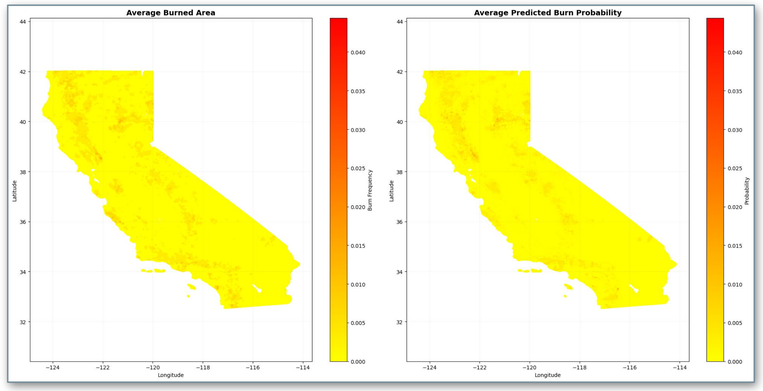
Baseline models struggled with the extreme class imbalance, often defaulting to predicting "no fire" for all samples.

Model	Prec (Fire)	Rec (Fire)	ROC-AUC	PR-AUC
Logistic Regression	*0.01*	*0.87*	*0.9287*	*0.0154*
Random Forest	*0.00*	*0.00*	*0.5325*	*0.0061*
Linear SVM	*0.00*	*0.00*	*0.9114*	*0.0468*

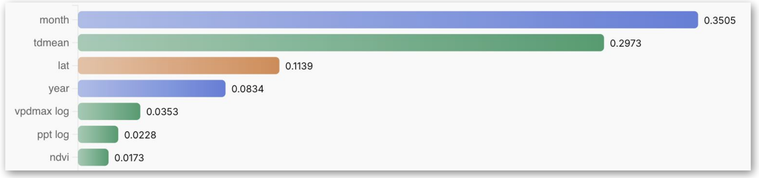
On the random test split, the tuned XGBoost model correctly identified 12,391 true fire events. When the model predicted fire, it was correct nearly 80% of the time. On temporally held-out data, the same model shows a much weaker minority-event signal.

XGBoost	0.60	0.38	0.937	0.31
Tuned XGBoost	0.78	0.43	0.9448	0.5548

This pattern suggests the model learns strong in-distribution ranking signal but remains sensitive to temporal shift. For planning workflows, the model is useful as a comparative risk-ranking tool, but it still requires calibration and threshold strategy before high-confidence deployment decisions.



Feature Importance



High maximum temperature (tmax) and vapor pressure deficit (vpdmax) are the strongest drivers of fire activity.

High precipitation (ppt) and vegetation moisture/greenness (NDVI) serve as the primary inhibitors of ignition and spread.

Modeling Methodology

Missing Data

▸ Missing values primarily occurred in **ocean and non-land grid cells**

Train / Test / Validation Strategy

▸ **Temporal validation set:** 2005–2009 held out for final evaluation
▸ Remaining years split:• **70% training 30% test**

Three Baseline Models

▸ Logistic Regression ▸ Random Forest ▸ Linear SVM

Final Model

▸ Gradient-boosted decision trees to improve rare-event detection.

XGBoost

▸ Handles nonlinear feature interactions
▸ More robust to extreme class imbalance
▸ Trained with **GPU acceleration** for scalability
▸ Tree depth, learning rate, subsampling ratios optimized using **Optuna**

Limitations

Compared to the strongest baseline, tuned XGBoost performs well on random splits but PR-AUC drops to 0.0113 on temporal validation

▸ Indicates sensitivity to nonstationary climate patterns and extreme class imbalance

▸ Current model should be interpreted as a research-stage risk signal, not an operational decision tool

Next Steps

Threshold & Probability Calibration

▸ Tune decision thresholds for rare-event prediction
▸ Calibrate predicted probabilities to reduce false confidence in wildfire risk

Temporal Robustness

▸ Implement stricter temporal cross-validation during model selection and tuning
▸ Improve model stability across different climate regimes
▸ Improve **temporal generalization**, since future-year holdout performance remains the biggest limitation of the current model

Feature Expansion

▸ Evaluate additional predictors related to ignition drivers (e.g., lightning, human activity)
▸ Explore cost-sensitive learning strategies to better capture rare wildfire events

*To **learn more** about our project, see relevant **prior literature**, and try our **interactive demo…***

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